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Polystyrene - PS

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Sep 4 2001

Polystyrene homopolymer, known as "crystal" polystyrene in the trade, is an amorphous, colorless, and transparent commodity thermoplastic. It is rigid, brittle, relatively hard and has excellent gamma radiation resistance, good electrical properties but has poor chemical and UV resistance. It can also be radiation sterilized.

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When modified by incorporating elastomers, polystyrene can be made into (high) impact polystyrene or HIPS, which is usually opaque. As a result, a wide variety of commercial polystyrene grades are available.

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Polymer Type

Thermoplastic

Advantages

Advantages of Polystyrene include:

- Cheap, rigid, transparent, easy to mould and good dimensional stability.
- Good electrical properties, low dielectric loss.
- Excellent resistance to gamma radiation.

Disadvantages

Disadvantages of Polystyrene include:

- (/)
 - Brittle, poor chemical resistance especially to organics.
 - Susceptible to UV degradation.
 - Flammable.

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Applications

Applications include toys, light diffusers, beakers, cutlery, general household appliances, video/audio cassette cases, electronic housings, refrigerator liners, housewares, containers, glazing, packaging and for HIPS.

Structural foam polystyrene moldings are used for business machine housings, tools, cases and boxes. Expanded polystyrene beads are used for packaging and cushioning. Foamed polystyrene is used for food trays, dishes, and egg boxes.

Typical Properties

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Value

Density (g/cm ³)	1.07
Surface Hardness	RM80
Tensile Strength (MPa)	34
Flexural Modulus (GPa)	3
Notched Izod (kJ/m)	0.02
Linear Expansion (1/°C x 10 ⁻⁵)	7
Elongation at Break (%)	1.6
Strain at Yield (%)	1.4
Max. Operating Temp. (°C)	50
Water Absorption (%)	0.05
Oxygen Index (%)	18
Flammability UL94	HB
Volume Resistivity (log ohm.cm)	16
Dielectric Strength (MV/m)	20
Dissipation Factor 1 kHz	0.0002
Dielectric Constant 1 kHz	2.6
HDT @ 0.45 MPa (°C)	90

HDT @ 1.80 MPa (°C)
Thermal Aging hr: @ (°C)
Mould Temp. Range (°C)
Mould Shrinkage (%)
Mould Temp. Range (°C)

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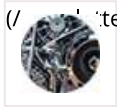
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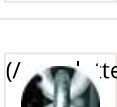
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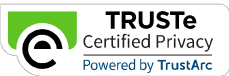
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